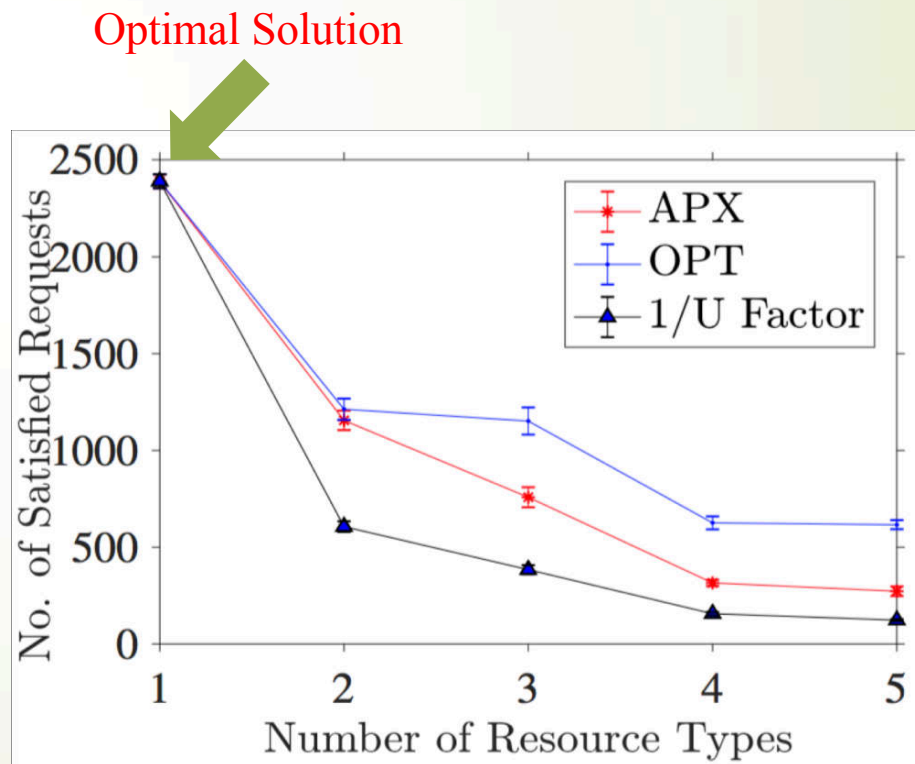
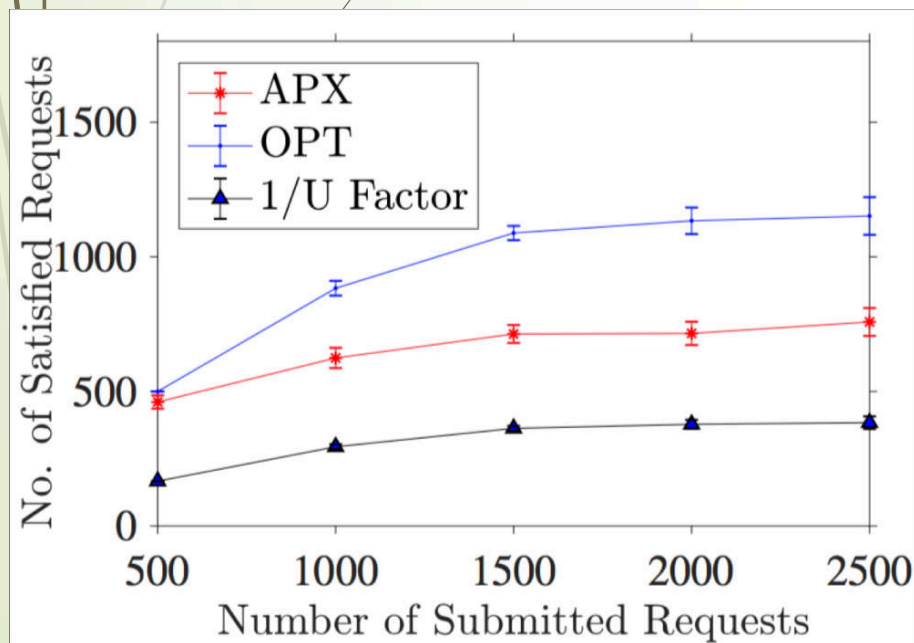


$\frac{1}{|U|}$ Approximation Factor (Sketch)

- 1: Design an upper bound solution (OPT')
 - Some simplifications: e.g., Sum up different resource types
 - 2: Prove results of OPT' $\geq$ optimal solution (OPT)
 - 3: Derive the approximation factor $\frac{1}{|U|}$ (APX/OPT')
- The algorithm results in optimal solution while $|U| = 1$

Validation

- APX is always above theoretical $1/U$ bound
- APX == OPT when $U=1$



Formulation #2 (Fewer Assumptions)

	Formulation #1	Formulation #2
Operator	Any Size	Predefined
Location	Grids	Any
Device Heterogeneity	Proportional	Any
Resource Constraints	Node	Node + Link

Problem Formulation

$$\max \sum_{q \in \mathbf{Q}} p_q \quad \text{Decision Variable} \quad (1a)$$

$$st : z_q = \sum_{i \in \hat{\mathbf{V}}_{\mathcal{A}_q}} \sum_{k \in \mathbf{V}_{q,i}} x_{q,i,k} / |\hat{\mathbf{V}}_{\mathcal{A}_q}| \quad \forall q \in \mathbf{Q}; \quad (1b)$$

$$z_q - 1 < p_q \leq z_q \quad \forall q \in \mathbf{Q}; \quad (1c)$$

$$\sum_{k \in \mathbf{V}_{q,i}} x_{q,i,k} \leq 1 \quad \forall q \in \mathbf{Q}, i \in \hat{\mathbf{V}}_{\mathcal{A}_q}; \quad (1d)$$

$$\sum_{q \in \mathbf{Q}} \sum_{i \in \hat{\mathbf{V}}_{\mathcal{A}_q}} \dot{F}(\cdot) x_{q,i,k} \leq R_{k,u} \quad (1e)$$

$$\forall k \in \mathbf{V}_i, u \in \mathbf{U};$$

$$\sum_{q \in \mathbf{Q}} \sum_{i \in \hat{\mathbf{V}}_{\mathcal{A}_q}} \sum_{i' \in \hat{\mathbf{V}}_{\mathcal{A}_q}} \sum_{k \in \mathbf{V}_{q,i}} \sum_{k' \in \mathbf{V}_{q,i'}} y_{q,(i,i'),(k,k')} \quad (1f)$$

$$J_{q,i,i'} T_{(k,k'),l} \bar{F}(\cdot) \leq B_l \quad \forall l \in \mathbf{E};$$

$$x_{q,i,k} = \sum_{k' \in \mathbf{V}_{q,i'}} y_{q,(i,i'),(k,k')} \quad (1g)$$

$$\forall q \in \mathbf{Q}, i \in \hat{\mathbf{V}}_{\mathcal{A}_q}, i' \in \hat{\mathbf{V}}_{\mathcal{A}_q}, k \in \mathbf{V}_{q,i};$$

$$x_{q,i',k'} = \sum_{k \in \mathbf{V}_{q,i}} y_{q,(i,i'),(k,k')} \quad (1h)$$

$$\forall q \in \mathbf{Q}, i \in \hat{\mathbf{V}}_{\mathcal{A}_q}, i' \in \hat{\mathbf{V}}_{\mathcal{A}_q}, k' \in \mathbf{V}_{q,i};$$

$$p_q, x_{q,i,k} \in \{0, 1\} \quad \forall q \in \mathbf{Q}, i \in \hat{\mathbf{V}}_{\mathcal{A}_q}, k \in \mathbf{V}_{q,i}. \quad (1i)$$

Objective

Constraints

Node

Link

SSE Heuristics (Algorithm)

Device
Selection

- Analytics selection Lower device load
 - Scarcest resource first
- Source/destination device selection Lower latency
 - Shortest path first
- Near-by device selection Lower network load
 - Early feature extraction

Polynomial-Time as well!

Evaluations

- Analytics (requests)
 - Poisson arrival / departure rates: 1 min / 10 mins
- Network topology (BRITE [1])
 - Number of devices: [10, 25, 50, 75, 100]
 - Device location and network bandwidth are from BRITE
- Available resources
 - CPU: [100% ~ 800%]
 - RAM: [1 GB ~ 16 GB]
- Bandwidth of links:
 - [45 kbps (LoRa), 8 Mbps (WiFi), 25 Mbps (4G)]

Baseline Algorithms

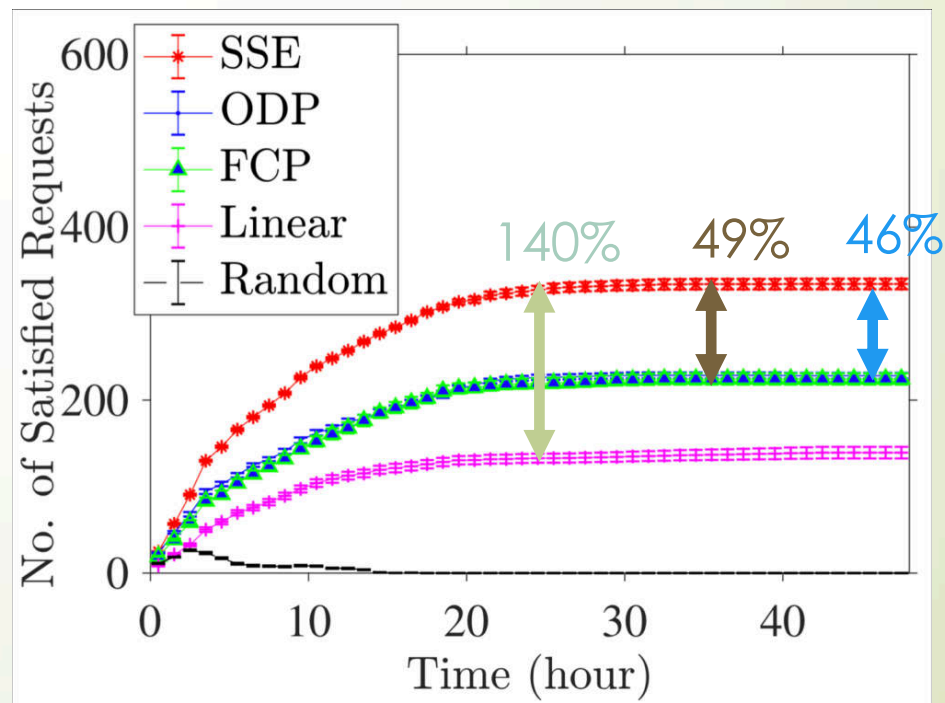
- Optimal Data Stream Processing Placement (ODP) algorithm [DEBS'16]
 - Commercial Solvers, e.g., CPLEX **State-of-the-art**
- Fog and Cloud Placement (FCP) algorithm [IM'17]
 - Heuristic
- Linear algorithm
 - Greedily deploys operators to neighboring devices
 - Considers all the constraints
- Random algorithm
 - Mimics a platform without centralized server
 - Does not consider any constraint

[DEBS'16] V. Cardellini, V. Grassi, F. Presti, and M. Nardelli. Optimal operator placement for distributed stream processing applications. In Proc. of ACM International Conference on Distributed and Event-based Systems (DEBS), Irvine, CA, June 2016.

[IM'17] M. Taneja and A. Davy. 2017. Resource aware placement of IoT application modules in Fog-Cloud Computing Paradigm. In Proc. of IFIP/IEEE Symposium on Integrated Network and Service Management (IM). Lisbon, Portugal.

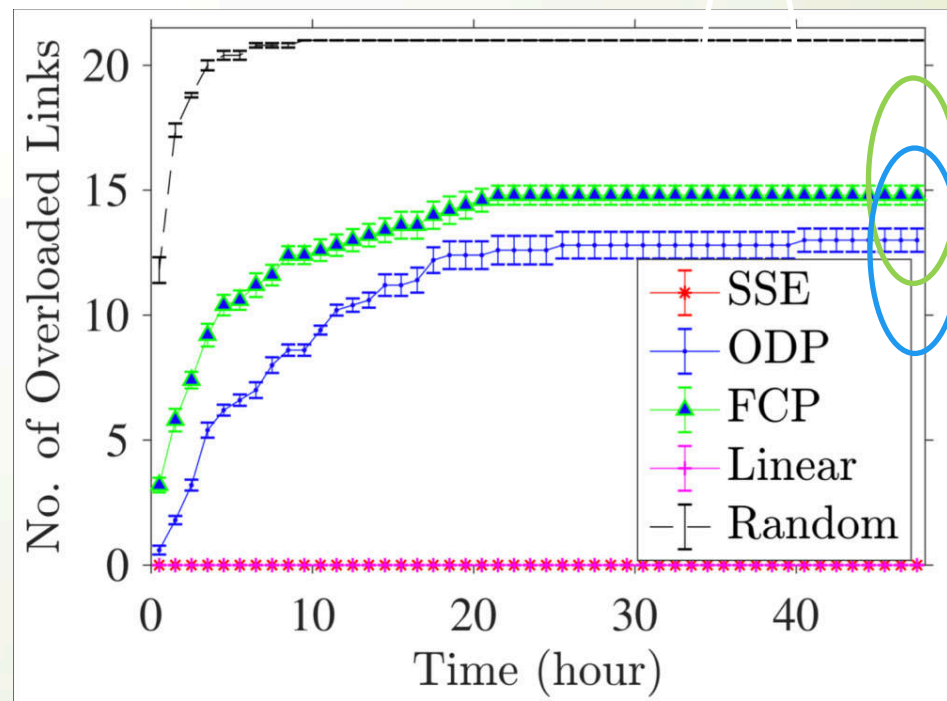
SSE Serves More Analytics Requests (While Achieving the QoS Requirements)

- 140%, 49%, and 46% compared to Linear, FCP, and ODP
- Random satisfies zero requests after running for 14 hours



Why SSE Serves More Analytics?

- Random, FCP, and ODP are too aggressive
- Overload links by up to 21, 12, and 10 links



SSE is (Also) Resource Efficient

- 6% and 12% more inactive devices (which can be turned off) compared to ODP and FCP
- Use less CPU/RAM/bandwidth to satisfy more requests

